

VAJRALA GEETHIKA

Software Engineer

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Summary

Aspiring Engineer graduate with a strong foundation in **Python, data structures, and object-oriented programming**. Skilled in developing efficient, scalable applications using **Python and Java** with a focus on clean, maintainable code. Experienced in collaborating within the **Software Development Lifecycle** to debug complex systems and implement test driven development practices. Committed to continuous learning and applying technical problem-solving skills to contribute effectively to software engineering teams.

Technical Skills

Programming & Databases:	Python (Advanced), Java, SQL (PostgreSQL, MySQL)
Software Engineering:	Data Structures & Algorithms, Object-Oriented Programming (OOPs)
Development Lifecycle:	Software Development Lifecycle (SDLC), Test-Driven Development, Debugging
Data Analysis & Stats:	Exploratory Data Analysis (EDA), Data Cleaning, Statistical Analysis
Machine Learning:	Supervised/Unsupervised Learning, Scikit-learn, Model Evaluation
Visualization:	Power BI (Dashboard Development), Advanced Excel (Pivot Tables, Lookups)
Tools & Concepts:	ETL Pipelines, Git/GitHub, Jupyter Notebook, VS Code

Projects

Loan Default Risk Prediction Pipeline | Python, Scikit-learn, Pandas

- **Architecture:** Engineered an end-to-end **Machine Learning Pipeline** to automate loan default risk assessment for a dataset of **50,000+ financial records**.
- **Feature Engineering:** Applied advanced normalization and feature engineering techniques, improving model sensitivity by **15%**.
- **Optimization:** Achieved a **0.82 ROC-AUC score** through rigorous hyperparameter tuning and cross-validation, significantly reducing potential credit loss.
- **EDA:** Conducted deep-dive statistical analysis to uncover latent borrower repayment patterns and visualize primary risk factors.

Customer Purchase Behavior Analytics | SQL, Python, Power BI

- **Data Engineering:** Analyzed 3,000+ transaction records using **Optimized SQL Queries** for complex data aggregations and trend discovery.
- **Business Impact:** Automated routine reporting tasks via Python Scripting, resulting in a **30% reduction** in manual data processing time (10 hours/week).
- **Visualization:** Architected interactive **Power BI dashboards** to track critical KPIs and product demand patterns, enhancing stakeholder decision-making speed.
- **Accuracy:** Improved data reliability for downstream modeling by **15%** through systematic data cleaning and outlier detection.

AI-Powered Computer Vision Attendance System | Python, OpenCV, SQL

- **Machine Learning Implementation:** Engineered an automated face-recognition system using **OpenCV** to manage high-volume records, increasing logging accuracy by **25%**.
- **Data Architecture:** Integrated a robust **MySQL** database backend to handle 1,000+ student profiles with sub-second retrieval latency and high data integrity.
- **Optimization:** Fine-tuned detection algorithms to maintain **98% accuracy** under varied environmental lighting conditions, ensuring system reliability.

EDUCATION

SASTRA Deemed University | Bachelor of Technology in Electronics and Communication Engineering
Thanjavur, Tamil Nadu

CGPA: 7.5 / 10

CERTIFICATIONS

- **Microsoft Certified:** Python for Data Science
- **Scaler Certified :**SQL & Database Design with MySQL